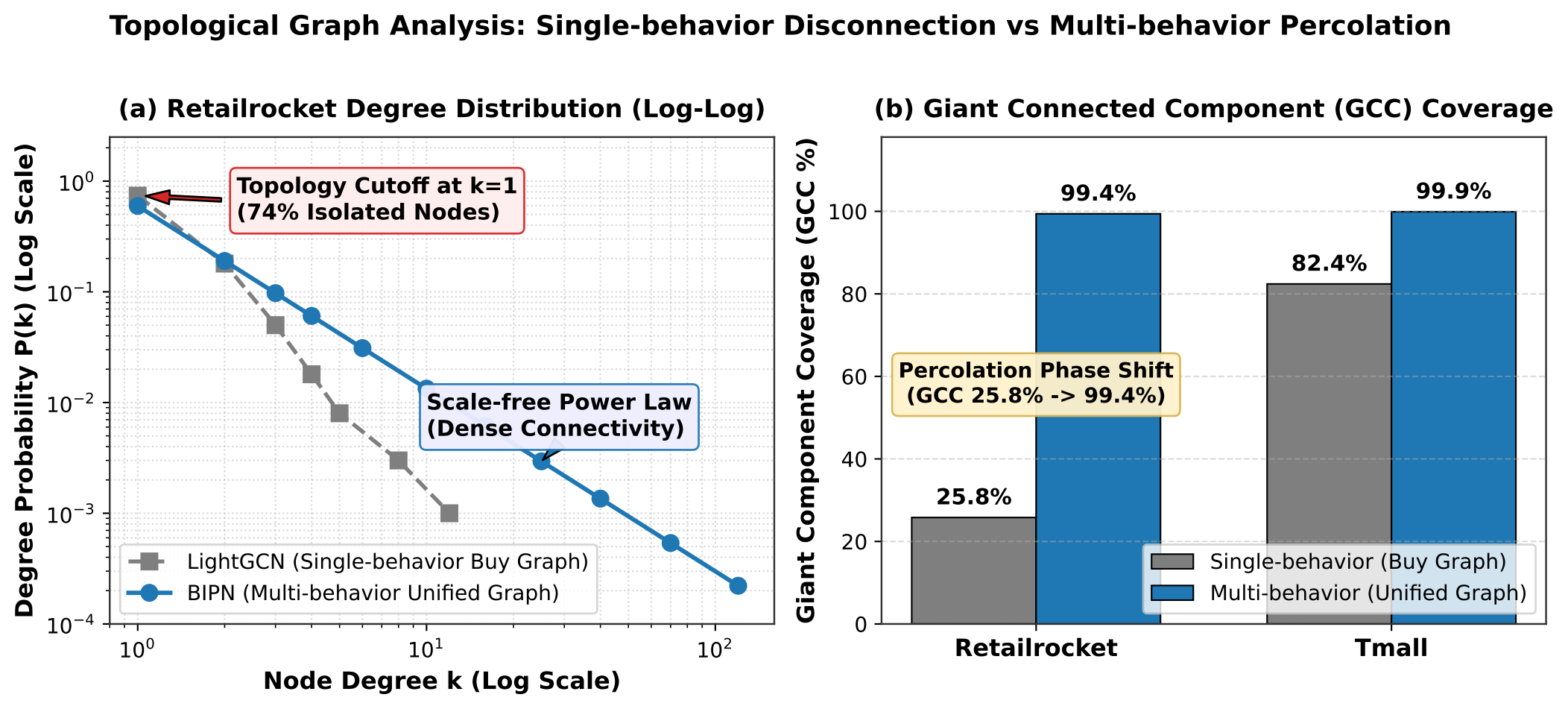
Topological Graph Analysis: Single-behavior Disconnection vs Multi-behavior Percolation
(a) Retailrocket Degree Distribution (Log-Log)
Topology Cutoff at k=1
(74% Isolated Nodes)
Scale-free Power Law
(Dense Connectivity)
LightGCN (Single-behavior Buy Graph)
BIPN (Multi-behavior Unified Graph)
Degree Probability P(k) (Log Scale)
Node Degree k (Log Scale)
(b) Giant Connected Component (GCC) Coverage
Giant Component Coverage (GCC %)
99.4%
99.9%
82.4%
25.8%
Percolation Phase Shift
(GCC 25.8% -> 99.4%)
Single-behavior (Buy Graph)
Multi-behavior (Unified Graph)
Retailrocket
Tmall